12.2.1 Normal Microbiota of the Skin

The skin's normal microbiota contains relatively large numbers of Gram positive bacteria, such as *Staphylococci* and *Micrococci*. Bacteria in the skin tends to be grouped into small clumps. Vigorous washing can reduce their numbers but will not eliminate them. Microorganisms remaining in hair follicles and sweat glands after washing will soon reestablish the normal populations. Areas of the body with high moisture, such as armpits and between the legs, have higher populations of microorganisms. They metabolize secretions from the sweat glands and are the main contributors to body odour.

Also part of the skin's normal microbiota are Gram positive pleomorphic rods called diphtheroids. Some diphtheroids, such as *Propionibacterium acnes*, are typically anaerobic and inhabit hair follicles. These bacteria produce propionic acid, which helps maintain the low pH of skin, generally between 3 and 5

12.2.2 Wound Infection

Wound can be defined as any interruption of continuity of external or internal surfaces caused by violence

Wounds may occur following: surgery, trauma or injections

Wound infections may occur mainly after surgical procedures

Wound sepsis is the result of cross infection from human sources and from other outside sources.

Bacteria associated with wound infections

Many bacteria are associated with wound infection (Figure 12.6). The normal flora may also cause infection. The most common normal flora of the skin are: *Staphylococci*, and various *Streptococci*, *Sarcina* sp, anaerobic Diphtheroids, Gram negative rods and others.

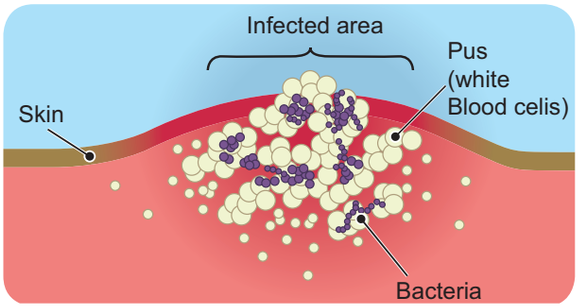


Figure 12.6: Bacterial infection on the skin

Post operative infections

Gasgangrene organisms like *Clostridium perfringens, Staphylococcus aureus* and *Clostridium tetani* may cause post operative infections.

HOTS

1. What are the possible infecting agent you could pick up when you are injured while playing on the ground? List them and name the diseases that they could cause.
2. What are the possible infectious agent that can infect you when you are injured by a rusted nail?

Route of entry

Wounds may occur following surgery, trauma or injections. Wound infections may occur mainly after surgical procedures. Wound sepsis is the result of cross infection from human sources and from other outside sources. Infections of skin are listed in Table 12.2.

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